

Engineered dual-scale poly (ϵ -caprolactone) scaffolds using 3D printing and rotational electrospinning for bone tissue regeneration

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ABSTRACT

Large bone defects due to trauma or disease present a significant clinical challenge with limited efficacy of current therapies. A key aim is to develop biomimetic scaffolds that reflect the native tissue structure with 3D printing being an important enabling technology. However, the incorporation of multiple length scales and anisotropic features, mimicking the native architecture, is difficult with current processes. In this study, we propose a simple and versatile hybrid printing process using a screw-assisted additive manufacturing technique combined with rotational electrospinning to fabricate dual-scale anisotropic scaffolds. 3D microscale porous polycaprolactone (PCL) structures with highly aligned nanoscale fibres were successfully produced layer-by-layer. The scaffolds were morphological, mechanical and biological characterised. Human adipose-derived stem cells (hADSCs) were seeded on the hybrid scaffold to evaluate the effects of structural and anisotropic topographic cues on cell attachment, proliferation and osteogenesis differentiation. Results show that the 3D printed microscale structures have uniform and well-defined geometries and the alignment of nanoscale electrospun fibres increases by increasing the electrospinning rotational velocity. Mechanical results show that there is no significant difference between 3D printed scaffolds with or without electrospun meshes. *In vitro* results show higher cell seeding efficiency and proliferation in dual-scale scaffolds with high density electrospun meshes. A more stretched and elongated cell morphology was observed in aligned nanofibre scaffolds showing higher anisotropic cytoskeletal organization than 3D printed PCL scaffolds without electrospun meshes. The dual-scale scaffolds present improved overall osteogenic markers expressions (COL-1, ALP and OCN). However, no statistical difference between normalised osteogenic marker expressions were observed between dual-scale scaffolds and 3D printed scaffolds. This might be attributed to the poor bioactivity of the substrate material, PCL, suggesting topographical cues might not be sufficient to stimulate cell fate towards to an osteogenic lineage. The results suggest that the proposed fabrication strategy is a promising approach for the design of novel bone scaffolds to modulate cell fates by integrating the topographic cue reported in this paper with biochemical cues associated to the use of more bioactive materials.

1. Introduction

Bone has significant capacity for regeneration and remodelling in cases of trauma and fracture but in events which induce a critically sized defect, bone can no longer heal by itself [1]. Moreover, this regenerative ability decreases with increasing age [2]. In these cases, clinical approaches are typically based on the use of autografts, the gold standard. Additionally, allografts and xenografts are also used. However, these approaches present poor clinical outcomes and have a

shortage of supply, potential transmission of disease, risk of rejection, and immunogenicity issues [3]. Tissue engineering solutions based on the use of bone scaffolds, combining advanced biomanufacturing strategies, synthetic and natural biomaterials, cells, and biomolecules, is a rapidly expanding multidisciplinary field which aims to repair, restore, and replace damaged or diseased tissues [4,5]. A bone scaffold is a biocompatible, biodegradable and mechanically stable substrate designed to mimic the native tissue and extracellular matrix (ECM) thus providing an environment for guide bone remodelling with minimal

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complications [6]. The success of this approach is dependent on a wide range of scaffold characteristics including material composition, morphology, surface properties, presence of biochemical cues, and biomechanics [7]. Current studies focus on exploring a variety of biomaterials such as bioceramics [8], natural/synthetic polymers [9], and metals [10].

However, there has been limited clinical success in scaffolds for bone regeneration due to the complexity of the native tissue. Bone is a hierarchical and anisotropic tissue with highly complex interconnected pore characteristics [5,11]. On the microscale, bone matrix consists of planar lamellae of aligned mineralised collagen with longitudinal alignment of bone cells [12]. Such alignment patterns effectively contributes to the osteogenic lineage of the mesenchymal stem cells [13]. On the nanoscale, the collagen fibrils are composed of aligned tropocollagen chains with rod or needle-like hydroxyapatite (HA) deposited within them. This gives rise to the remarkable mechanical properties of bone. Therefore, it is of great interest to design and fabricate a dual-scale anisotropic scaffold which can mimic the structural and anisotropic properties of native bone tissue.

Traditional fabrication methods have been utilised to incorporate anisotropic and multiscale features in tissue engineering scaffolds. Bae et al. [14] used replica moulding and surface wrinkling to fabricate poly (urethane acrylate) (PUA) micro/nano substrates and showed that NIH3T3 fibroblast cells are more adapted to the multi-scale surface pattern in terms of morphology and orientation. Cui et al. [15] fabricated ECM-like micro/nano fibrous scaffolds of HA/poly(lactide-co-glycolide) using melt electrospinning. Results show that the ECM-like structure promotes bone mesenchymal stem cells proliferation and differentiation *in vitro* and a distinct bony callus of bridging was observed after 12 weeks of *in-vivo* studies. Maleki et al. [16] fabricated silica-silk fibroin bioaerogel scaffolds using a sol-gel reaction, unidirectional freeze-casting and supercritical drying approach. The aerogel presented a hierarchical structure with ordered honeycomb-like micromorphology and multiple length pore characteristics. *In vitro* studies showed that the hybrid gel can trigger human osteosarcoma MG-63 cell attachment and proliferation. Results also showed significant bone tissue density formed in the defect sites after 25 days of implantation. These studies present promising results in terms of using multi-scale structures for bone tissue engineering. However, traditional manufacturing techniques, with the associated complexity, have limited capability to design scaffolds with specific and reproducible morphologies (eg. pore size, porosity, and interconnectivity), hierarchical length-scales, and the incorporation of multiple materials including biomolecules and cells.

Additive manufacturing can overcome many of these limitations by allowing the fabrication of scaffolds with precisely designed morphologies which can incorporate multi-material structures [17]. Furthermore, the crystal volume fraction and the level of anisotropy in 3D printed thermoplastic filaments can be controlled by controlling processing conditions in screw-assisted additive manufacturing [18]. However, the resolution of commonly used additive manufacturing, in particularly extrusion-based additive manufacturing (the most common technology for bone scaffolds for tissue engineering applications) is typically limited to the microscale [19]. To mimic the nanoscale fibres of the ECM alternative technologies are required such as electrohydrodynamic (EHD) processes (eg. electrospinning, electrospraying, and melt electrospinning) which are capable of producing sub-micron fibres [20–22]. Typically, traditional electrospinning produces randomly aligned fibres, however, the use of rotating collecting mandrels, parallel substrates separated by a gap, magnetic substrates, and patterned collectors enable the fabrication of uniaxial aligned arrays of nanofibres [23–25]. Furthermore, 3D printing deposition principles have been used to facilitate EHD jet printing and melt electrospinning writing to produce highly aligned sub-micro fibrous dual-scale scaffolds [26–28]. However, scaffold thickness is limited due to charge dissipation and the process can be time consuming to produce highly dense

and ordered structures. The combination of additive manufacturing and electrospinning is a promising approach to design and produce multi-scale scaffolds allowing to start mimicking the complexity and hierarchical structure of bone. Several studies reported the use of a combination of additive manufacturing and electrospinning to fabricate hybrid scaffolds [29–33]. However, produced electrospun meshes are typically highly dense with randomly distributed fibres, placed directly on top of the scaffold or the use of inter-dispersed meshes in the scaffold. Furthermore, the knowledge of osteogenic behaviour of hybrid 3D printed and electrospun scaffolds incorporating multiple length scales and aligned electrospun fibres is limited. A key challenge to modulate osteogenic behaviour is the design of a hybrid scaffold with optimised electrospun mesh density and degree of alignment of nanofibres to promote a specific response (eg. osteogenic differentiation).

This study uses a hybrid fabrication process combining screw-assisted extrusion-based additive manufacturing and rotational electrospinning, providing a simple and versatile platform to fabricate multi-material and multi-scale structures. This enables the creation of an integrated dual-scale scaffold with specifically designed electrospun mesh densities and nanofibre alignment. A biocompatible polymer, polycaprolactone (PCL), is used for both the 3D printing and electrospinning. This allows understanding of the fabrication process and how mesh density and fibre alignment can be engineered. The mechanical properties are investigated to assess any changes in the dual-scale structure. The biological behaviour including cell attachment, proliferation, morphology, and osteogenic differentiation of human adipose derived stem cells (hADSCs) is characterised as a function of multiple length scales, mesh density and fibre alignment.

2. Materials and methods

2.1. Materials

PCL was obtained from Perstorp Caprolactones (UK). Fetal bovine serum (FBS), Dulbecco's modified eagle medium (DMEM), phosphate buffered saline (PBS), hexamethyldisilazane (HDMS), 10 % neutral buffered formalin, Triton X-100, Alizarin Red-S (ARS), 10 % ammonium hydroxide, radioimmunoprecipitation assay (RIPA) buffer, anti-collagen type I primary antibody and protease inhibitor cocktail were all purchased from Sigma-Aldrich (UK). Human adipose-derived stem cells (hADSCs), MesenPRO RS™ basal media, STEMPRO™ osteogenesis differentiation kit, Alexa Fluor 488-conjugated phalloidin, 4',6-diamidino-2-phenylindole (DAPI), goat anti-mouse IgG secondary antibody (Alexa Fluor Plus 647) and the bicinchoninic acid assay (BCA, Micro BCA Protein Assay Kit) were all obtained from Thermo Fisher Scientific (UK). Sensolyte®pNPP Alkaline Phosphatase (ALP) assay kit was purchased from AnaSpec (USA). MTT cell proliferation and osteocalcin (OCN) assays were obtained from Abcam (UK). Human RUNX2 (Runt-related transcription factor 2) enzyme-linked immunosorbent assay (ELISA) kit was obtained from FineTest (China).

2.2. Fabrication

Dual-scale scaffolds were produced using a screw-assisted extrusion additive manufacturing system (3D Discovery, RegenHU, Villaz-Saint-Pierre, Switzerland) combined with a rotating drum electrospinning system (Spraybase, Ireland)(Fig. 1a). To enable the combined fabrication process, a 3D printed mandrel enclosure and removable platform were fabricated and wrapped with aluminium foil to ensure conductivity of the components (Fig. 1b). The scaffold was fabricated on the removable platform and fixed in place with printing tape and then moved manually between the 3D printer and rotational electrospinner. The platform was fixed into a holder on each machine to ensure correct positioning. Microscale 3D structures were printed using 0°/90° lay-down pattern, 350 µm pore size and 270 µm layer thickness, melting temperature of 90 °C, screw rotational velocity of 12 rpm and 20 mm/s

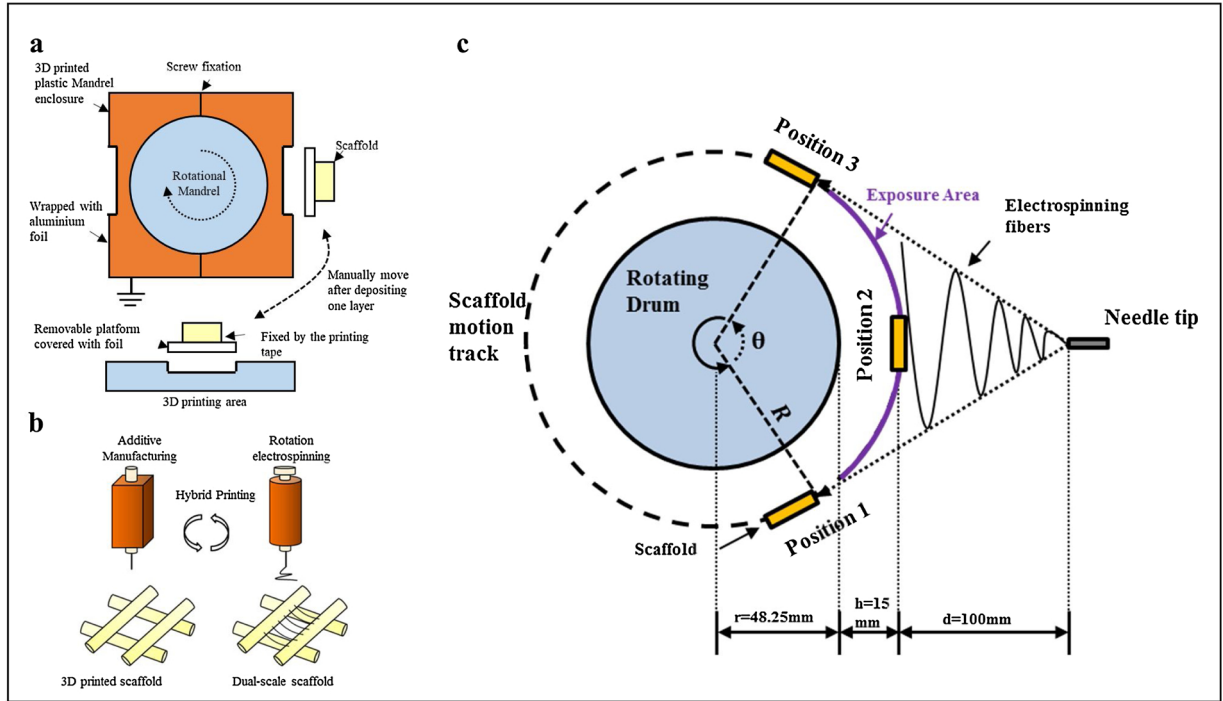


Fig. 1. Schematic representation showing (a) the hybrid printing processes (b) experimental set up and (c) rotational electrospinning process and corresponding geometric configurations.

deposition velocity. After printing two initial layers, the scaffold was then electrospun using the rotational electrospinning system with a variety of drum rotational velocities (0 rpm, 100 rpm, 500 rpm and 1000 rpm) to create low and high alignment of the electrospun fibres (Fig. 1a). This procedure was repeated till the final fabrication of the scaffold. The PCL solutions for electrospinning were prepared by dissolving PCL (20 % w/v) in glacial acetic acid and acetone (70/30 % v/v), whilst being stirred at 40 °C overnight. The electrospinning processing parameters used were a flow rate of 1 mL/h, 100 mm needle-collector distance, 17 kV charge, and 1 mm inner diameter needle. Electrospun fibre mesh density was controlled by changing the electrospinning time. In the case of electrospinning without rotation (non-rotational electrospinning), two different electrospinning times (45 and 120 s) were considered to fabricate low and high density meshes. A schematic representation of the fabrication process is shown in Fig. 1c. In the case of rotational electrospinning, the scaffold was electrospun at position 1 and finalise at position 3 in each cycle of rotation. As the electrospinning time refers to the time the scaffold is electrospun, it was necessary to determine the equivalent electrospinning time (T_e) for rotational electrospinning in order to obtain meshes with similar density in the case of non-rotational electrospinning. The exposure time in each cycle, T_{exp} , can be calculated as:

$$T_{exp} = \theta / w \quad (1)$$

where,

$$\theta = 2 \cdot \arccos(R/L) \quad (2)$$

and

$$w = k \cdot R_{pm} \quad (3)$$

In these equations, θ is the angle of exposure area (in radians), R is the radius of the scaffold motion track (the distance from the top layer of the scaffold to the centre of the drum), L is the distance between the needle tip and the centre of the drum, w is the angular velocity, k is a constant ($2\pi/60$) and R_{pm} is the rotational velocity. In order to obtain the same exposure time as in the non-rotational electrospinning case

(T_{ran}), the equivalent electrospinning time for rotational electrospinning T_e can be obtained by:

$$T_e = R_{ev} / R_{pm} \quad (4)$$

where R_{ev} , corresponding to the number of cycles to obtain the equivalent electrospinning time, is given by:

$$R_{ev} = T_{ran} / T_{exp} \quad (5)$$

Thus, the equivalent time, T_e , can be determined as follows:

$$T_e = k \cdot T_{ran} / \theta \quad (6)$$

According to Eq. (6), the equivalent time depends on the non-rotational electrospinning time and the experimental geometry lay-out. In this study, the considered rotational electrospinning times were 2 and 6 min to produce low and high density meshes. According to the mesh density and drum rotational velocity, samples were named as shown in Table 1.

2.3. Morphology

Morphological analysis was performed by scanning electron microscopy (SEM) (TESCAN VEGA3, TESCAN, UK). Scaffold morphology

Table 1
Sample types and their processing parameters.

Samples	Electrospun mesh	Electrospinning processing time (s)	High/Low density mesh	Rotational velocity (rpm)
PCL	N/A	N/A	N/A	N/A
PLR	Yes	45	Low	N/A
PHR	Yes	120	High	N/A
PL100	Yes	120	Low	100
PH100	Yes	360	High	100
PL500	Yes	120	Low	500
PH500	Yes	360	High	500
PL1000	Yes	120	Low	1000
PH1000	Yes	360	High	1000

and fibre dimensions were analysed using Fiji software and fibre orientation was analysed using the polymer network software SOAX following the procedures described by Xu et al. [34]. The pore size of electrospun fibres was measured on the SEM images using the DiameterJ plug-in within the Fiji software [35]. The generated XY coordinates of the nanofibres were used as input data into the MATLAB software (MathWorks, USA) to calculate the fibre orientation with respect to the y-axis of each SEM image using a vector analysis of the adjacent identified fibre coordinates. The analysed data was plotted using the Origin software (OriginLab Corporation, USA). The same protocol was used to determine cell fibril orientation. In this last case, confocal fluorescence images of cells were used rather than SEM images.

2.4. Mechanical analysis

Compression tests were performed using a universal testing machine (Instron 3344, Instron, Massachusetts, USA). Scaffolds ($n = 7$) were cut into $3 \text{ mm} \times 3 \text{ mm} \times 4 \text{ mm}$ cubes and tested in a dry state at a strain rate of 0.5 mm/min with a 100 N load cell. The compressive modulus was obtained from the slope of the stress-strain curve and the apparent yield stress was obtained by the intersected point of the parallel line to the elastic gradient with a strain offset of 0.2% .

2.5. Biological analysis

In order to assess the effects of multiple length scales (micro and nanofibre), mesh densities, and aligned nanofibres on cellular functions, PCL, PLR, PHR, PL1000 and PH1000 scaffolds ($6.5 \text{ mm} \times 6.5 \text{ mm} \times 2.5 \text{ mm}$) were considered for biological assessment.

2.5.1. Cell culture and seeding

Scaffolds ($n = 4$) were sterilised with 80% ethanol for 4 h and rinsed with PBS at least 3 times, followed by drying overnight. The hADSCs (passage 4) were cultured using MesenPRO RS™ basal media supplemented with 2% (v/v) growth serum, 1% (v/v) glutamine, and 1% (v/v) penicillin/streptomycin and were seeded onto the scaffold with a cell density of $20,000$ cells per sample. The osteogenic differentiation media was prepared using an osteogenesis basal medium supplemented with a 10% (v/v) osteogenic serum and 1% (v/v) penicillin/streptomycin. Osteogenic media was added to the cell seeded scaffolds after 24 h of culturing in growth media to allow initial cell attachment. Subsequently, all differentiation assays at time points of 3 , 7 , 14 and 21 days are equivalent to a total culture time of 4 , 8 , 15 and 22 days. Cell culture medium and differentiation medium were changed every 3 days.

2.5.2. Cell attachment and proliferation

MTT assay was used to assess the initial cell seeding efficiency and proliferation after 4 , 72 , and 168 h . Briefly, the culture media was removed. Scaffolds were moved to a new plate and $150 \mu\text{L}$ of DMEM and $150 \mu\text{L}$ of MTT reagent were added into each well. After 3 h of incubation at 37°C , the solution was removed and $450 \mu\text{L}$ of MTT solvent was added. After shaking in the dark at 37°C for 15 min , the absorbance was measured at 590 nm using a microplate reader (Infinite 200, Tecan, Männedorf, Switzerland).

2.5.3. Osteogenic differentiation

ALP activity was used to evaluate cell differentiation at an early stage of osteogenesis (day 3 , 7 , and 14). The experimental procedures followed the manufacturer's instruction. Briefly, the scaffolds ($n = 4$) were washed twice with $1\times$ alkaline phosphatase dilution assay buffer and lysed with a 0.2% (v/v) Triton X-100 solution in $1\times$ assay buffer. The samples were vortexed for 1 min followed by sonication for 3 min and then freeze-thawed twice at -80°C . The samples were then centrifuged at $2500 \times g$ (relative centrifugal force) for 15 min at 4°C followed by the extraction of the supernatant which was mixed with the

para-nitrophenyl phosphate (pNPP) substrate and incubated for 1 h at room temperature. The absorbance was measured at 405 nm using a microplate reader. The concentration of ALP was calculated based on a standard curve and normalised to the total protein concentration which was determined by using the BCA.

The level of RUNX2, an early-stage osteogenic marker, was determined at day 3 , 7 , and 14 through ELISA. Samples were rinsed with PBS twice to remove residual osteogenesis media and lysed in RIPA cell lysis buffer containing a protease inhibitor in a shaker for 30 min at 4°C . The cell lysis solution was centrifuged at $3500 \times g$ for 15 min at 4°C . Following the manufacturer instructions, $50 \mu\text{L}$ of the supernatant was diluted with $50 \mu\text{L}$ of the dilution buffer and added to the pre-coated 96 well-plate and incubated for 90 min at 37°C . $100 \mu\text{L}$ of a biotin-labelled antibody solution was added and incubated for 60 min at 37°C and $100 \mu\text{L}$ of horseradish peroxidase (HRP) streptavidin conjugate solution was subsequently added. After incubation for 30 min at 37°C , samples were visualised by adding $90 \mu\text{L}$ of $3,3',5,5'$ -tetramethylbenzidine (TMB) substrate to react with the HRP for 20 min at 37°C which was stopped by adding $50 \mu\text{L}$ of an acidic stop solution. The absorbance was measured at 450 nm by a microplate reader. The concentration of RUNX2 was calculated by obtaining an interpolated value from the standard curve and multiplying by the dilution factor. The obtained values were normalised to the total protein concentration.

A late stage osteogenesis marker, OCN, was detected at day 21 using a similar protocol to the RUNX2 measurement. Scaffolds ($n = 4$) were lysed, cell lysis solution centrifuged, and the supernatant added into a pre-coated 96 -well microplate followed by the addition of $50 \mu\text{L}$ of the antibody cocktail. After incubation for 1 h at room temperature, the samples were washed three times, mixed thoroughly with $100 \mu\text{L}$ of the TMB substrate solution and shaken in the dark for 10 min . After adding $100 \mu\text{L}$ of stop solution, the samples were read using a microplate reader at an absorbance of 450 nm . The concentration of OCN was obtained by interpolation of the standard curve and normalising to the total protein concentration.

The mineralisation of the scaffolds was evaluated using ARS staining at day 21 . The scaffolds ($n = 4$) were rinsed with PBS solution twice to remove media and fixed using 10% formalin for 15 min . Scaffolds were then washed three times with deionised water and stained with a 0.2% (w/v) ARS solution for 40 min . The scaffolds were then rinsed with deionised water five times to remove unbound ARS. A 10% acetic acid solution was added with gently shaking for 30 min . The solution was removed and heated up to 80°C for 10 min , cooled down in ice for 15 min , and centrifuged at $20,000 \times g$ for 5 min . Finally, $400 \mu\text{L}$ of supernatant was neutralised with $80 \mu\text{L}$ of 10% ammonium hydroxide and were read at an absorbance of 405 nm using a microplate reader.

2.5.4. Cell morphology

Cell morphology was analysed using SEM and confocal microscopy (SP8, Leica Microsystems, Germany) at day 1 and 3 , respectively.

For SEM imaging, scaffolds were fixed with 10% formalin for 30 min , dehydrated using a series of increasing concentrations of ethanol and finally HMDS, and then allowed to dry overnight prior to imaging.

Confocal microscopy was used to determine cell morphology and expression of collagen type I (COL-1). Briefly, the samples were fixed with 10% formalin for 30 min , rinsed with PBS three times, and permeabilised using 0.2% Triton X-100 solution for 10 min . Samples were washed with PBS three times and 5% FBS solution was added to block non-specific binding for 60 min at room temperature. Anti-collagen type I primary antibody ($1:400$ dilution) in 1% FBS solution was added to each sample and incubated overnight at 4°C . The secondary antibody ($1:400$ dilution) in 1% FBS solution was then added and incubated for 1 h at room temperature in the dark. Alexa Fluor 488-conjugated phalloidin ($1:400$ dilutions) with 1% FBS solution was added and incubated for 45 min in the dark. Finally, cell nucleus staining was performed by adding $1 \mu\text{g/mL}$ of DAPI and incubating for 5 min in the

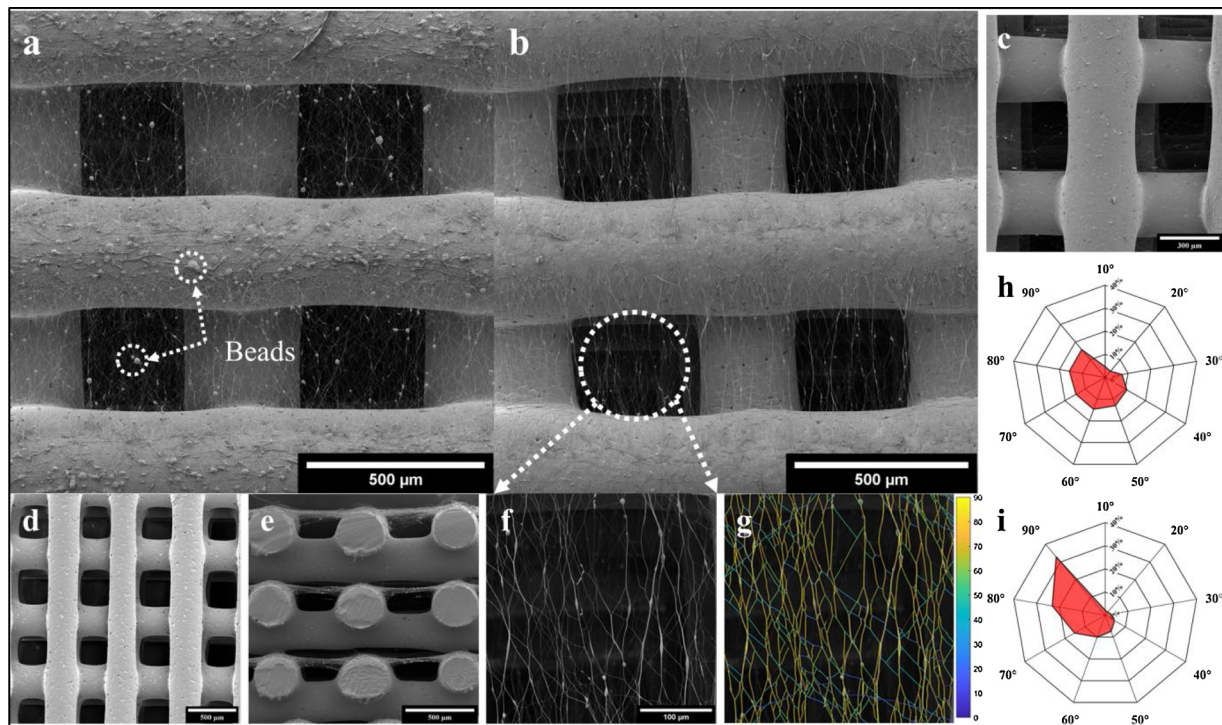


Fig. 2. SEM images of (a) PHR scaffold; (b) PH1000 scaffold; (c) PLR scaffold; (d) PCL scaffold; (e) cross-section of PHR scaffold; and (f) high magnification SEM image of PH1000 scaffold; (g) the fibre orientation analysis of PH1000 scaffold (colour bar unit: °); (h) the radar plot of PHR scaffold showing the percentage of fibres in all direction; (i) the radar plot of PH1000 scaffold showing the percentage of fibres in all direction.

dark. The samples were then analysed by confocal microscopy and the anisotropy of fibril was investigated following the procedures reported by Boundaoud et al. [36] by randomly selecting 30 locations and analysing using the FibrilTool function in Fiji software. Cell numbers were counted in accordance to DAPI staining using the Fiji software.

2.6. Statistical analysis

Statistical analysis was performed using one-way analysis of variance (ANOVA) with Tukey post-hoc test in GraphPad Prism software (Graphpad Software Inc., San Diego, CA, USA). Differences were considered significant at $*P < 0.05$, $**P < 0.01$, $***P < 0.001$ and $****P < 0.0001$.

3. Results

3.1. Morphology

The dual-scale scaffolds were successfully produced using a combination of screw-assisted extrusion-based additive manufacturing and electrospinning (Fig. 2 and Supplementary Fig. S1). The results show that the 3D printed microscale structures are well-defined and have morphological features within the designed parameters (filament diameter: $315.3 \pm 25.7 \mu\text{m}$, filament pore size: $348.3 \pm 22.1 \mu\text{m}$, layer thickness: $271.7 \pm 60.8 \mu\text{m}$). The pores and filaments in the 3D printed scaffold are uniformly distributed and the nanoscale electrospun meshes are deposited onto alternative 3D printed layers thus having a uniform direction within the scaffold (Fig. 2d and e). The nanoscale electrospun fibres are mostly below $1 \mu\text{m}$ (Table 2). The pore size of electrospun fibres decreases by increasing the electrospinning time, indicating that the mesh density increases by increasing the electrospinning time (Fig. 2a and c and Table 2). Although the electrospinning time was adjusted to minimise differences between rotational and non-rotational electrospinning, the mesh density for non-rotational electrospinning is higher than the density of meshes produced using

rotation. This can be attributed to the difficulty in controlling the deposition of electrospun fibres onto the scaffold during rotational electrospinning, as the scaffolds only interact with the fibre jet for a short period of time during each rotational cycle whilst in the non-rotational process the interaction is continuous. This difference has been attempted to be controlled by increasing the electrospinning time (2 and 6 min), however, discrepancies remain, and further optimisation is required. The electrospun fibres deposited on the 3D printed filaments connect parallel printed filaments by bridging the filament gap (pore), forming an integrated hierarchical dual-scale system. It is possible to observe that the electrospun fibres have a degree of alignment in all sample types and this anisotropic characteristic is enhanced using rotational electrospinning and increasing the rotational velocity (Fig. 2b, f and g and supplementary Fig. S1). Fig. 2h and i and Table 2 provide numerical results, showing that the orientations of PHR electrospun fibres are homogeneously distributed in the region between 40° and 90° while for PH1000 most fibres (more than 50 %) are in the direction between 80° and 90° . The use of high rotational velocities during electrospinning also contributes to the changes observed on the surface properties of both the micro printed filaments and nano electrospun fibres. The electrospun meshes produced with no rotation and low velocity (100 rpm) show the presence of beads both in the mesh within the pores and on the printed filaments, furthermore, the fibres have a lower degree of alignment on the 3D printed filament surfaces (Fig. 2a and supplementary Fig. S1). However, these effects are minimised using high rotational velocities (500 and 1000 rpm).

3.2. Mechanical analysis

The mechanical properties of the dual-scale scaffolds are shown in Fig. 3. The results indicate no statistical differences between PCL scaffolds and dual-scale scaffolds in terms of both the compressive modulus and apparent yield stress. Furthermore, mesh density seems to have no significant effect on mechanical properties either. This might be attributed to the thin layer of electrospun fibres and the relatively

Table 2
Nanofibre mesh characteristics in the dual-scale scaffolds.

	PCL	PLR	PHR	PL100	PH100	PL500	PH500	PL1000	PH1000
Fibre orientations (%)									
10°	–	5.9	4.6	5.0	3.4	2.5	4.8	2.3	2.2
20°	–	5	3.5	7.1	2.6	3.1	3.6	0.5	1.7
30°	–	6.2	7.4	7.8	3.8	2.5	5.1	0.7	2.6
40°	–	9.3	10.8	11.6	5.5	3.6	6.0	0.6	4.7
50°	–	11.1	12.9	7.6	8.4	6.4	8.0	1.4	7.4
60°	–	13.1	14.7	7.8	12.4	11.2	9.8	3.5	10.0
70°	–	16.7	14.5	12.5	17.6	17.6	12.5	6.5	15.3
80°	–	17.8	15.7	18.4	22.5	24.8	19.7	30.2	23.2
90°	–	14.9	15.8	22.0	23.8	28.3	30.4	54.2	32.9
Fibre diameter (nm)	–	752 ± 263	741 ± 197	925 ± 237	969 ± 478	747 ± 187	729 ± 256	969 ± 463	817 ± 397
Pore size (µm ²)	–	178 ± 655	30 ± 110	360 ± 860	83 ± 244	144 ± 389	32 ± 104	353 ± 669	140 ± 508

small volume of the electrospun meshes compared to the 3D printed structure.

3.3. Biological analysis

3.3.1. Cell attachment and proliferation

Cell attachment and proliferation results of the hADSCs are shown in Fig. 4. The PHR scaffolds present significantly higher cell seeding efficiency (~80 %) than the other scaffold types. PLR, PL1000 and PH1000 scaffolds present similar seeding efficiency but, in all cases, higher than the PCL scaffolds. This can be attributed to the presence of electrospun fibres which increases the effective surface area for cell attachment and facilitates cell bridging between 3D printed filaments. The non-rotational electrospun mesh forms denser structures which captures cells more effectively compared to the rotational electrospun fibres which are more aligned resulting in larger pores thus cells can slip through these gaps. This can be observed in Fig. 4d and e showing cells attached to both printed filaments and electrospun fibres and more cells are attached to the non-rotational electrospun mesh. In all scaffolds, the number of cells increases with time (Fig. 4b), showing no cytotoxicity effects. Results show that, at the same time point, PHR scaffolds present higher cell number than PLR, PL1000 and PH1000 scaffolds and statistically significantly higher cell numbers than PCL scaffolds. Moreover, cell morphology seems to be influenced by the surface topography of the scaffolds. As shown in Fig. 4c–e, the cell cytoskeletal seems to be elongated and more orientated with the aligned electrospun fibres in PH1000 scaffolds compared to PCL and PHR scaffolds, showing that the alignment of electrospun fibres has a significant effect on the cell anisotropic characteristics.

3.3.2. Cell differentiation and morphology

Fluorescence confocal microscopy was performed at day 3 of cell differentiation to enable cell morphology analysis (Fig. 5). Cells are observed attached to the scaffold with changes in their morphology in response to different surface topography. Cells attached to the 3D printed filament surfaces seem to grow along the printing direction (Fig. 5a–c and h). The cell cytoskeletal structures in PHR and PH1000 scaffolds are stretched and elongated due to the presence of the nanoscale electrospun fibres (Fig. 5d–g). Cells are more aligned along the direction of the electrospun fibres in the PH1000 scaffolds (Fig. 5f and g). The fibril orientation analysis (Fig. 5j and k) also show that the value of fibril orientation is increasing in the range of angles from 50° to 10°, showing more aligned cytoskeleton in the direction of electrospun fibres, while in case of PHR scaffolds the fibril orientations are more isotropically distributed. Fig. 5m also confirms that there is an increasing trend in terms of cell alignment anisotropy due to the presence of nanoscale electrospun fibres and PH1000 show higher anisotropy than other scaffolds, being the results statistically significant than PCL scaffold. This might be attributed to the multi-scale topographical structures and anisotropic alignment of fibres, acting as physical cues influencing cell mechanotransduction.

The presence of COL-1 is predominately co-localised with the cell nuclei, suggesting that hADSCs are initialising collagen synthesis (Fig. 5i). Fig. 5l shows the semi-quantification of COL-1 for all samples. The number of pixels representing COL-1 staining increases with the number of cells. Normalising the total number of pixels representing COL-1 to the number of cells results show that PHR scaffolds present higher secretion of COL-1 per cell than the other scaffolds.

The influence of the dual-scale scaffolds on osteogenic differentiation of hADSCs was assessed with specific assays for markers of early and late stage differentiation at days 3, 7, 14 and 21. An early stage

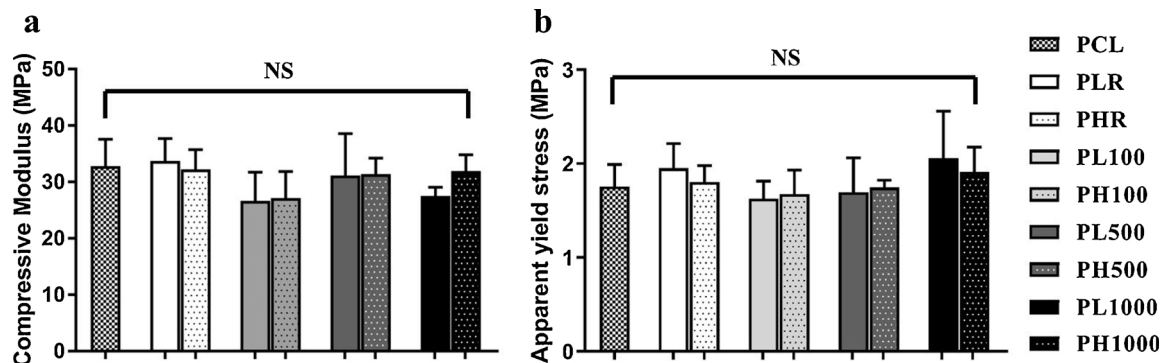


Fig. 3. (a) Compressive modulus and (b) the apparent yield stress for all scaffolds (NS: no significant difference).

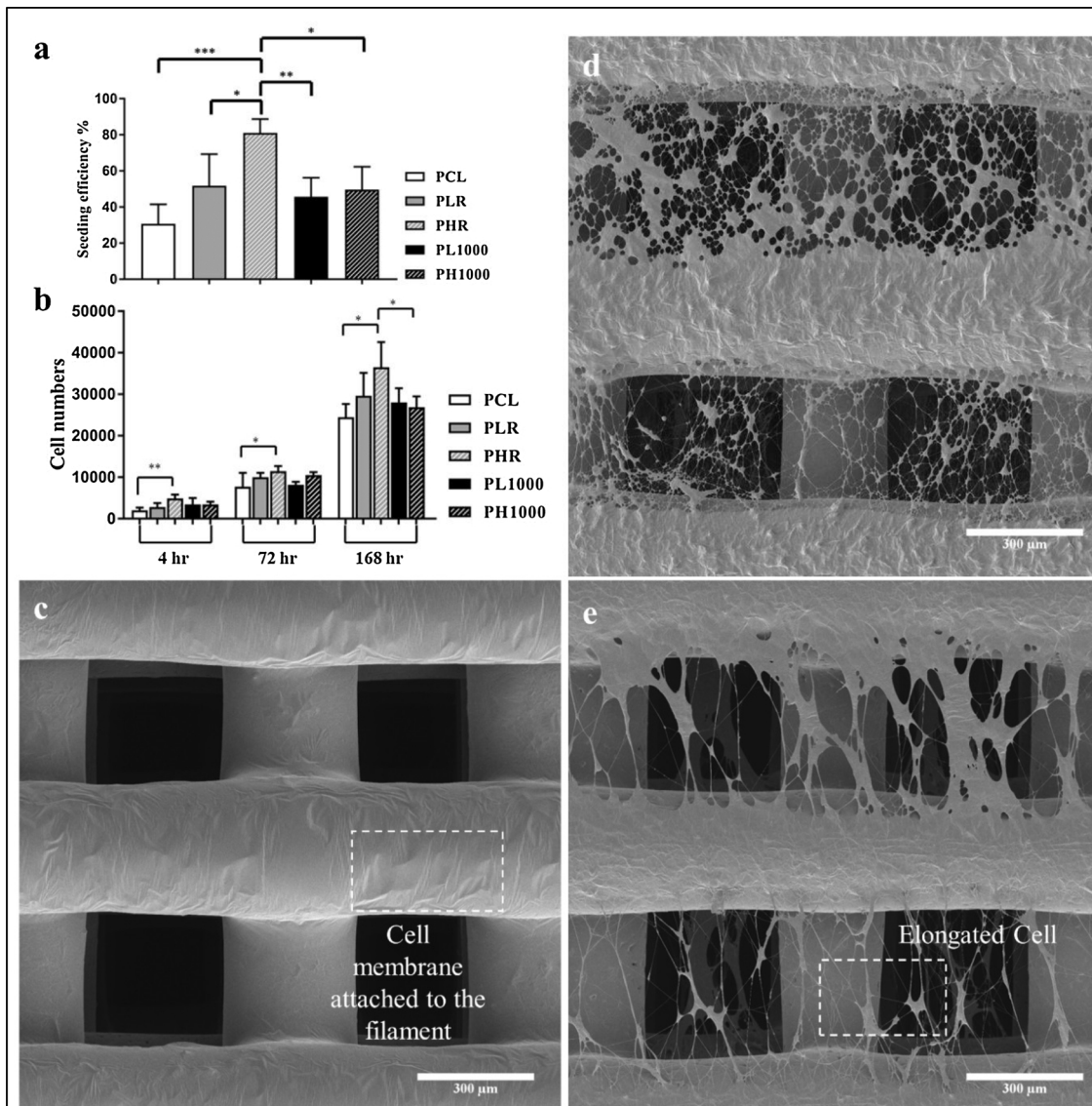


Fig. 4. (a) Cell seeding efficiency of all scaffolds after 4 h incubation; (b) MTT results showing the cell numbers at 4, 72 and 168 h; SEM images of (c) PCL scaffold, (d) PHR scaffold, and (e) PH1000 scaffold after 1 day of cell culture.

osteogenesis marker, RUNX2 protein, was assessed at days 3, 7 and 14 and the results are presented in Fig. 6a and b. Fig. 6a shows that the normalised RUNX2 protein levels for all scaffolds decreases with increasing the osteogenic differentiation time. No statistically significant differences were observed between samples. The results seem to indicate that normalised RUNX2 protein secretion slightly decreases with increasing mesh density and mesh anisotropy (Fig. 6a). A similar trend is observed for the overall RUNX2 protein secretion as indicated in Fig. 6b showing a decrease in protein expression with the increase of mesh anisotropy, but the overall RUNX2 protein expression increases from day 7 to day 14.

The ALP activity, an early stage osteogenesis marker, shows an opposite trend in terms of protein expression. Results show that both normalised ALP activity (Fig. 6c) and overall ALP expression (Fig. 6d) increased with increasing differentiation time. No statistically significant differences were observed at day 3. However, at day 7 PL1000 scaffolds exhibit statistically significant higher normalised ALP and overall ALP than PHR scaffolds. At day 7, it also seems that the ALP activity increases with the increase of electrospun fibre alignment. A similar trend is also observed at day 14, with results showing increased ALP activity for PH1000 scaffolds, suggesting that increased alignment

of the electrospun fibres positively affects ALP secretion.

The results of OCN, a late stage osteogenic differentiation marker, are shown in Fig. 6e and f. The normalised OCN concentration is higher in scaffolds containing electrospun fibres. However, there is no statistically significant difference between each sample. Fig. 6f shows that the overall OCN protein secretion for both PLR and PHR scaffolds are significantly higher than for PCL scaffolds. However, there is no significant difference between PL1000 and PH1000 scaffolds.

Scaffold mineralisation was determined through ARS staining and the results are shown in Fig. 6g. As observed, PCL scaffolds present slightly higher quantity of calcification than other scaffolds, however, no statistical differences were observed between samples.

4. Discussion

Bone is a hierarchical and anisotropic tissue comprising a network of cells, ordered arrays of mineralised tropocollagen fibrils and associated macromolecules [11]. Previous studies reported that nanoscale surface patterns promote the differentiation of mesenchymal stem cells towards osteogenic lineages [37] and the substrate's surface with continuous micropattern topography lead to increase of both cytoskeletal

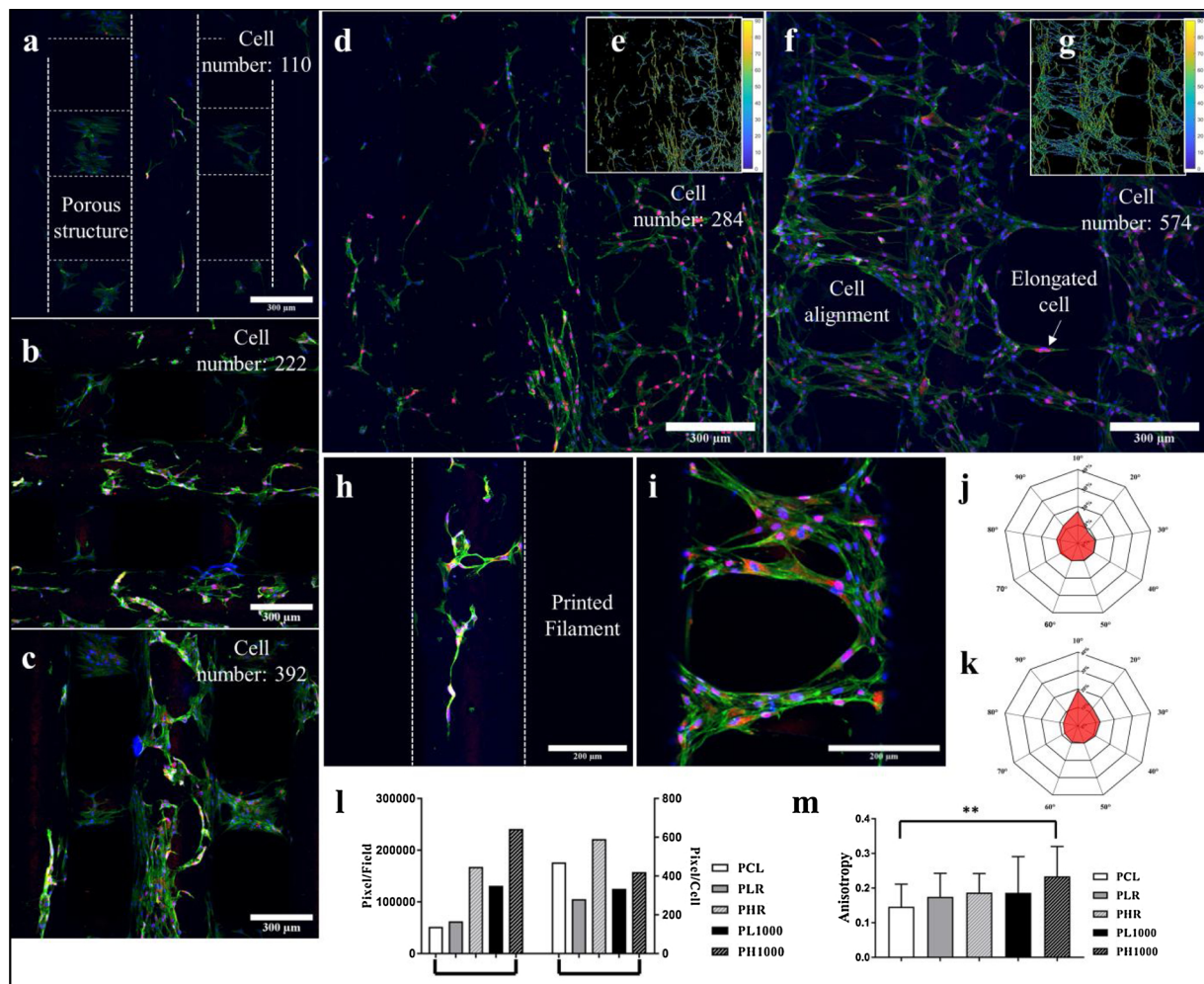


Fig. 5. Confocal images at day 3 showing (a) PCL scaffold, (b) PLR scaffold, (c) PL1000 scaffold, (d) PHR scaffold, (e) fibril orientation analysis for PHR scaffold (colour scale bar unit: °), (f) PH1000 scaffold, (g) fibril orientation analysis for PH1000 scaffold (colour scale bar unit: °), (h) high magnification image of printed single filament in PCL scaffold, (i) high magnification image of PH1000 scaffold; the radar plots of fibril orientation on (j) PHR scaffold and (k) PH1000 scaffold; (l) semi-quantification of collagen type I staining (left group to the left axis (pixels/field), right group to the right axis (pixel/cell)); (m) anisotropy quantification analysis for all scaffolds (blue: nuclei staining, green: actin staining, red: collagen type I staining).

anisotropy and osteogenic marker expression [38]. Therefore, it is highly relevant to design and fabricate synthetic bone grafts with hierarchical structures and anisotropic features.

In this study, a hybrid fabrication process integrating screw-assisted additive manufacturing with electrospinning (rotational and non-rotational) was developed to fabricate dual-scale anisotropic scaffolds. The hybrid printing process combines the advantages of both additive manufacturing and electrospinning. This enables the fabrication of precise 3D scaffolds with uniformly distributed filaments and pores. Furthermore, the incorporation of nanoscale electrospun meshes mimics the ECM thus increasing the effective surface area for cell attachment, promoting cell bridging between printed filaments, and modulating cell morphology. Several studies have reported the fabrication of dual-scale scaffolds [29–32]. However, in these cases, the fabricated meshes are highly dense which can inhibit cell penetration and scaffold permeability whilst the randomly distributed fibres on the scaffolds can fail to provide specific cell guidance. Gao et al. [39] developed a multi-scale direct writing system to produce multi-scale scaffolds with controlled filament/fibre deposition using an electrostatic field. However, this method is time consuming and difficult to mass produce aligned nanoscale fibres and the effects of ordered surface topography towards cellular behaviour, in particularly osteogenesis behaviour, were not investigated. The hybrid printing-electrospinning

process reported in this paper is a simple and effective method to produce dual-scale scaffolds with highly aligned nanoscale fibres. The use of rotational electrospinning is critical since it can not only produce highly aligned fibres, with higher alignment produced at higher rotational velocities, but can also influence the electrospun fibre characteristics on the printed filament. As shown in Fig. 2a, the use of non-rotational electrospinning produces low oriented electrospun fibres within the pores, but the fibres are randomly deposited on the printed filament surface and beads are present. This finding is supported by results previously reported by Li et al. [23,24] which attribute the disordered fibre distribution in spatial orientation to electrostatic interactions. However, the use of rotational electrospinning minimises this effect (Fig. 2b), leading to high orientation electrospun fibres crossing over the printed filaments and reduction in beads. Moreover, this phenomenon is enhanced by increasing the rotational velocity of the drum (supplementary Fig. S1), which decreases the cyclic exposure time of the electrospun fibres interacting with the printed filaments, leading to a reduction of beads and increases fibre alignment due to high velocities/mechanical stretching forces [40]. Mechanical analysis shows that the addition of electrospun fibres and the increase of mesh density have limited effects on the compressive modulus and mechanical strength. Mechanical properties such as stiffness can influence cellular responses including the regulation of osteogenic differentiation

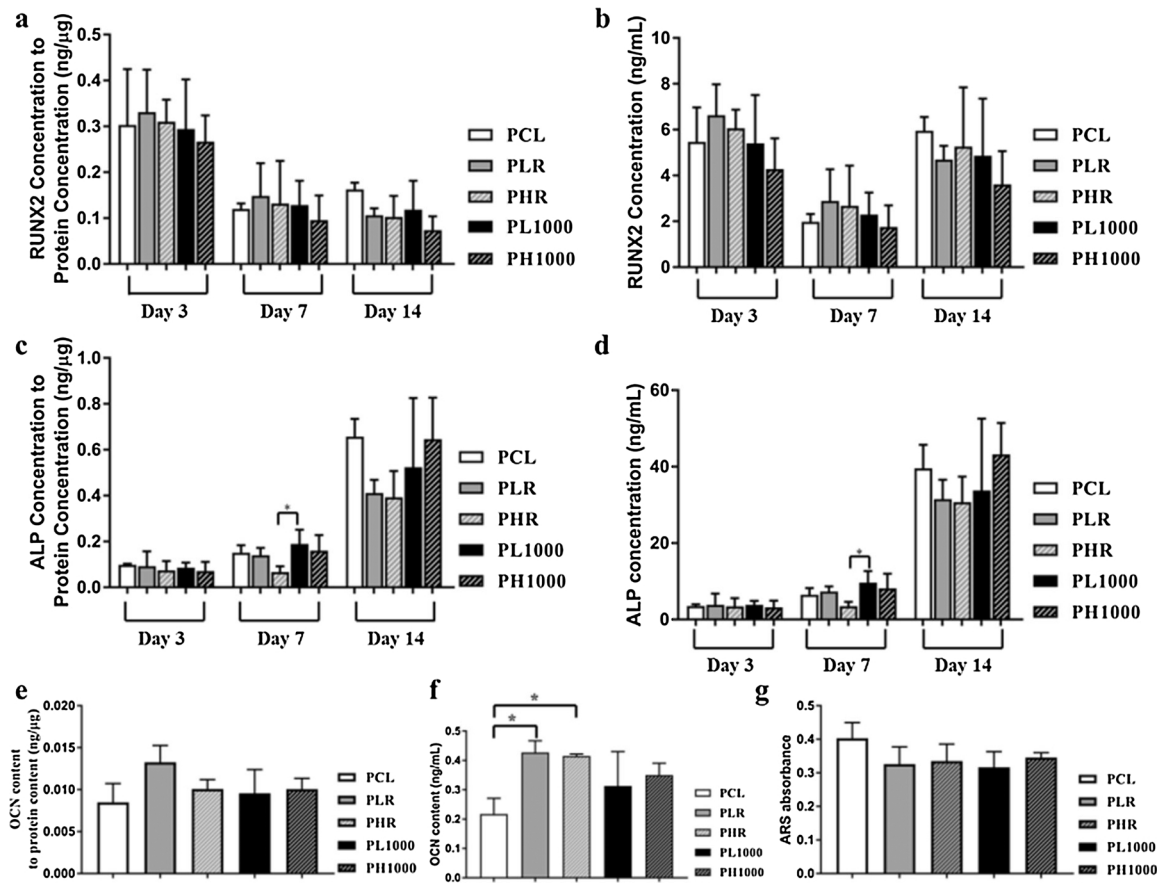


Fig. 6. Osteogenic differentiation of the dual-scale scaffolds. (a) Normalised RUNX2 and (b) total expression for all scaffolds at day 3, 7 and 14; (c) the normalised ALP and (d) total expression for all scaffolds at day 3, 7 and 14; (e) the normalised OCN and (f) total expression for all scaffolds at day 21; (g) the ARS results for all scaffolds at day 21.

of mesenchymal stem cells [41]. However, this is not a key contributing factor in this study since all scaffolds were produced using the same material, PCL.

Cells are sensitive to the biophysical and biochemical micro-environment and strategies to harness these will contribute to cell fate decisions, such as cell attachment, proliferation and differentiation [42,43]. A major contributing factor to cell fate is surface topography through both the specific physical structures and presentation of biochemical cues [13,44]. This study demonstrates that both the dual-scale structure and the aligned electrospun fibres topography influence hADSC behaviour. Considerably more hADSCs are attached to the dual-scale scaffolds compared to the 3D printed only scaffold which can be explained by the larger surface area of the dual-scale scaffolds. Increased cell migration and colonisation of the scaffold could also explain the higher cell proliferation observed for PHR scaffolds. As cell motility and migration speed can be significantly enhanced by using denser substrate surfaces and reduced substrate size [45]. The PHR scaffold presenting nanoscale and denser meshes allows faster cell migration contributing to closer cell-cell interactions within the scaffold. The ordered organisation of electrospun fibre mesh influences cell morphology. The aligned electrospun meshes in the PH1000 scaffold seem to guide the cell growth along the electrospun fibre direction resulting in elongation and stretching of the cell cytoskeleton. Similar results were observed by He et al. showing that cells culture on aligned nanofibers have an aligned cell morphology [46,47]. The dual-scale scaffolds exhibit significant increase in cell attachment and proliferation, especially the PHR scaffold, which enables scaffolds to be rapidly colonised by cells. This could provide a considerable benefit for tissue engineering applications, for example, in acellular scaffolds using a

dual-scale strategy could result in higher rates of cell infiltration and new tissue formation *in vivo*.

The multiple length scales and anisotropy of scaffolds can also impact osteogenic differentiation. As observed in this study, micro-scale PCL scaffolds present an overall more “stable” osteogenesis behaviour than hybrid scaffolds. Previous studies show that stem cells attached to microscale structures can cause a cascade of events through enhanced integrin activation, which triggers BMP2 signalling pathways by increasing SMADs protein expression, and eventually up-regulating the transcription factor RUNX2 which promotes ALP, COL-1 and OCN protein expression [48,49]. However, these studies also point out that the nanoscale substrates stimulate cell-cell communications through gap junctions by increasing connexin-43 (gap junction alpha-1 protein) expression, which can also stimulate RUNX2 expression promoting osteogenesis behaviour, suggesting the synergistic effects of using both micro and nanoscale hybrid structures instead of single length scale structures [48]. This may explain why some osteogenic markers (ALP and OCN expression) are higher in the dual-scale scaffolds than in the 3D printed PCL only scaffolds. Moreover, the anisotropic properties of the dual-scale scaffolds seem to be another factor that stimulates osteogenesis. As reported in this study, the use of rotational electrospinning increases the alignment of nanofibres, increasing the anisotropy of the differentiated hADSCs contributing to a more organised cell skeletal distribution. The expression of osteogenesis markers, in particular ALP, is improved by increasing the alignment of electrospun fibres. This study shows some evidence that the use of micro/nano hybrid structures and anisotropic nanopatternography (electrospun fibres) can promote osteogenic behaviour. However, no statistical differences between dual-scale and PCL scaffolds were observed. This can be attributed to the

poor bioactivity of the PCL, which was selected as an initial reference material for this proof of concept study as it has been extensively used by our group for bone tissue engineering applications [50–53]. This indicates that the presence of aligned electrospun fibres is not sufficient by itself to be a critical factor in promoting osteogenic behaviour. However, the dual-scale scaffold with features of rapid recruitment of cells and directional guidance of cell growth provides a unique hierarchical anisotropic micro/nano-environment for the large population of cells and induce increased overall osteogenic marker expressions (ALP, COL-1 and OCN). The osteogenic behaviour of the designed dual-scale scaffolds could be improved by using bioactive biomaterials and composites such as bioceramics (e.g. HA and bioglass) and natural polymers (e.g. collagen, gelatin, fibrin). Additionally, the combined 3D printing and rotational electrospinning provides a simple and versatile platform to fabricate multi-material and multi-scale structures. Materials and processing parameters can be changed *in situ* to incorporate different biomaterials and morphologies (e.g. fibre diameter, pore size, and surface topography) to mimic native tissue composition and architecture.

5. Conclusions

Dual-scale anisotropic PCL scaffolds were successfully fabricated using screw-assisted extrusion-based additive manufacturing integrating either rotational or non-rotational electrospinning. The 3D printed microscale structure is well organised with controllable morphological features which are embedded with nanoscale electrospun meshes that show increasing alignment with increasing rotational velocity of the drum collector. The developed dual-scale scaffolds and 3D printed only scaffold share similar mechanical properties, but the surface topography is significantly changed by increasing the electrospinning time and the rotational velocity. *In vitro* studies show that the use of high mesh densities increases cell seeding efficiency, allowing higher cell-cell interactions thus promoting cell proliferation and faster colonisation of the scaffold. The use of aligned nanofibres leads to more elongated and stretched cell morphologies compared to the low orientated nanofibres and microscale 3D printed filaments. Moreover, the use of aligned nanofibres in the 3D printed scaffolds can significantly promote the anisotropic cell organization, allowing more organised cytoskeleton. Aligned nanofibre scaffolds show higher expressions of osteogenic markers (COL-1, ALP and OCN) but no statistically significant differences were observed between scaffolds. The osteogenic differentiation could be significantly enhanced by using more bioactive materials and integrating the topographic cue reported in this paper with biochemical cues. This study presents a promising approach in the design of new scaffold based strategies to modulate cell attachment, proliferation, and morphology for bone tissue engineering applications.

CRedit authorship contribution statement

Boyang Huang: Conceptualization, Methodology, Software, Writing - original draft, Investigation, Formal analysis, Visualization, Supervision, Writing - review & editing. **Enes Aslan:** Conceptualization, Methodology, Writing - original draft, Investigation, Formal analysis, Visualization, Writing - review & editing. **Zhengyi Jiang:** Methodology, Software, Investigation, Formal analysis, Writing - review & editing. **Evangelos Daskalakis:** Investigation, Writing - review & editing. **Mohan Jiao:** Investigation, Writing - review & editing. **Ali Aldalbahi:** Conceptualization, Writing - review & editing, Funding acquisition. **Cian Vyas:** Conceptualization, Methodology, Software, Investigation, Supervision, Writing - review & editing. **Paulo Bártolo:** Conceptualization, Methodology, Validation, Resources, Supervision, Project administration, Writing - review & editing, Funding acquisition.

Declaration of Competing Interest

There is no conflict of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.addma.2020.101452>.

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